Biology

Chapter 3:
Organism and Their Environments

Welcome to AnithUncommon's **Sample Resources!**

We are assuming you are reading this as a student. So What will **YOU** Expect From This Sample?

In Biology, Organism and Their Environments , Ch. 3.1

This sample will be divided into **three** parts: Introduction to Samples (this part), Lesson Objectives (What is Expected From Students), and most crucially, Syllabus Contents.

Note that the 'Syllabus Content' does NOT stand as a complete syllabus itself; rather, it is a tight compilation of notes gathered by Azalia to create a comprehensive sample for students.

If you want a complete version of these notes and think that you need help in Biology, join our nonprofit!

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This sample is made thanks to Azalia.

— Made by students, for students.

Lesson Objectives:

At the end of the lesson, students should be *able* to...

1. **Define** the terms “population”, “community”, “ecosystem” and “Biosphere” as a part of **ecology levels**.
2. **Distinguish** between different **types of population** based on their **distribution patterns**.
3. **Analyzing** the graph of population growth, including **J-Shaped Graph** and **S-Shaped Graph.**
4. **Explain** the **factors** affecting population growth, including the roles of **density-dependent** factors and **density-independent** factors.

Reflective Question to Consider:

A population is not just about a collection of humans, animals, or bacteria, it’s an entire system that breathes, grows, and responds to its environment. These groups of organisms do follow a specific pattern of expansion and decline.

Below is the following core questions you will face as we explore this chapter:

- To what extent do density-dependent factors act as a balancer to the population’s stability?
- How does the distribution pattern of a population reveal the resources available in the habitat?

Unit 3: Organism and Their Environments

Pro Tip!

- **In this chapter, we will basically learn to be more detailed with our environments. All things here are about grouping, analyzing patterns, and making pyramids of level. That's three basic concepts that will help you in understanding this chapter.**

Activities in Class:

- **Lesson breakdown**
- **Blooket-Based review for deep memorizing**
- **Visualization of graphs using Draw Chat**
- **Examining real-life example of organism interaction**

3.1 Population

Five Levels of Ecology

A. What Are Those?

Ecology is a branch of science that studies **how living organisms interact with each other** and with the **environment** they're in. Talking about ecology levels, there are **five main levels** from **smallest** to the **most thorough: individuals, population, community, ecosystem, and biosphere.**

B. Definition of Ecology Levels

a. **Individuals**

Individual stands for a **single organism from a specific species**, like human (*Homo sapiens)* or a butterfly from *Danaus plexipus* species. The point

Mentor's Notes ★

When it comes to putting ecology levels in order, I use the "Zebra and Deer" analogy:

One zebra→ Individual
Other zebras coming to it → population
A bunch of deer asking for company to those zebras → Community

is: when we're **on our own,** we are individuals. b. **Population** As a social being, we won't grow up only by ourselves. We need to **interact** with other humans in the area near us. This situation, where **a group of individuals** under the same species **reside and interact in a specific area**, it's called population. c. **Community** When population consists only of those in the same species, community is when **2 or more species live and interact in a certain region.** d. **Ecosystem** We only talk about the living (biotics), but the environment also consists of the **abiotic components,** such as **water**. When a **biotic community lives alongside the abiotic ones,** supporting each other, then this creates an ecosystem. e. **Biosphere** The **whole ecosystem that exists on earth, merged into a single cavity** is called the biosphere. **Population Structure & Pattern of Distribution** **A. Quantifying Populations** "Quantifying" here means putting **the amount of individuals in a population into math logic.** It's all to understand **how 'crowded' a population is** before we predict **if they might run out of food** and elselike. There are two measurement concepts in this context: a. **Size of Abundance**	**They live together within the sun rays and the river → ecosystem** **The whole earth containing ecosystems → biosphere**

→ The **total number of individuals** in the population.

→ Ex: **"There are 290 cats in this neighborhood"**

b. **Density**

→ While size of abundance counts all individuals in the population one by one, density uses **individual amounts in units of area.**

$$\mathit{Density} = \frac{\textbf{\textit{Total Individuals}}}{\textbf{\textit{Total Space}}}$$

Answer example: ***57 plum trees per square kilometer.***

→We need to count density as in ecology, we don't just count size but also **competitions happen between organisms** in a certain area.

B. Dispersion Patterns

The pattern describes how **individuals in a population spread** across their habitat. Beside telling us **the place individuals settle**, they also tell us about the **behavior of certain individuals** and where **resources in that area** are positioned.

There are three kinds of patterns:

a. **Clumped**

→ Means **individuals gather in small groups**, like patches. This is the **most common pattern** found in nature.

→ Usually appears in a habitat where the

Think of density as a way to reveal the internal competition naked eyes observation don't see:

→ If 100 lions live in a teeny-tiny 1 km forest, logically, they will fight for food. This resembles the high density of lions in the area.

→ But what if the same 100 lions live in a super huge 1000 km savanna? They might get dispersed and never see each other again. This resembles a low density of lions in that area.

About distribution patterns, it is better

resources (like water) are only **found in specific spots**. Besides, this also **resembles the "togetherness"** in some animals' behavior, especially to get **protection**.
→ Ex: A pack of wolves.

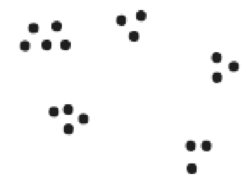
b. **Uniform**
→ Means **individuals** in that population, somehow, **spaced themselves evenly** between each other.
→ Usually as a result of **territoriality.** Individuals develop a specific "bubble" to **ensure they have enough space and food**.
→ Ex: When penguins breed, they lay eggs on desert bushes that release a poison for other plants that wanna grow close to them.

c. **Random**
→ Means **Individuals scattered with no pattern** we can conclude. In fact, this rarely happened in nature.
→ Appears in a habitat where **resources are plentiful all across** the place. The **individuals** living there **are solitary** so they don't form groups as the clumped type does.
→ Ex: Dandelions have a wind-dispersed seed that will follow anywhere the wind goes, and mostly land on random fields, adding up the diversity there.

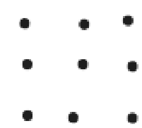
Here's the summary in table:

Pattern	Visualization	Main Reason
Clumped	Clusters,	Uneven

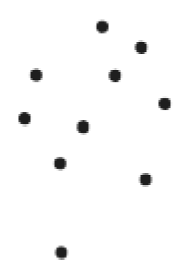
to make visualizations, such as:



Clumped ones will look like this. They clumped in where the resources are.



And this grid, is how the uniform looks like. Yes, they're alone, defending their territories in an even space between each other.



This one? So random, right? Just like the name, it has

	group	resources or social bonds (most common)
Uniform	Grid-like	Territory rules
Random	No pattern	Even resources (very rare)

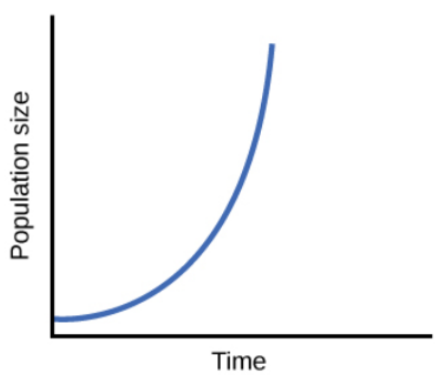
Dynamics of Population Growth

A. Growth Models

The fact that **"nothing lasts in this world"** is exactly the reason why populations are **dynamically growing**. In this topic, we will sum the whole story of a population through a single line going up and down.

Basically, we use graphs to **describe and measure the dynamics** of a population. There are **two types of population graphs**:

a. **The J-Shaped Curve (Exponential)**



Look at the upward graph above. This type of graph tells us that the **population growth** is **unlimited and rapid** since the **resources are infinite** and the habitat **has no predators**.

→ This graph is **rare** in nature, since every habitat and place has **finite resources** that

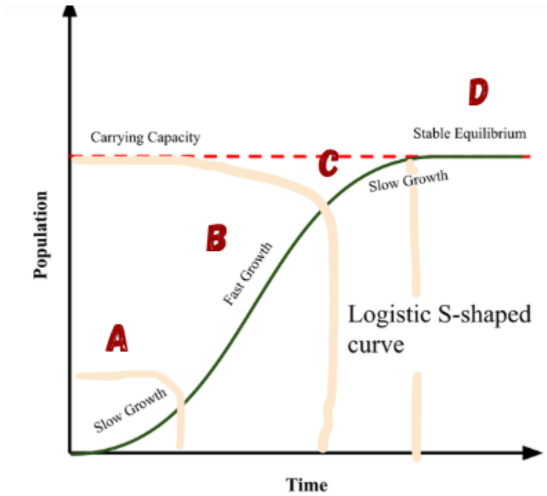
no patterns so it will look like that.

The J-Shaped Curve is usually only applicable in the lab, not in nature since it resembles the "perfect growth" no population could do.

Just remember if you see this kind of graph, it means the growth is limitless since there are no brakes or lines.

can't support infinite growth of population years to come.

b. **The S-Shaped Curve (Logistic)**



This S-looking graph depicts the real-world situation of how a population grows. It has four phases to note on:

1. **Phase A: Lag Phase**
   The region in the **bottom left** of the graph means that the **growth is slow** because there are not **many parents** to reproduce.
2. **Phase B: Log Phase**
   This wide region means that the **population explodes** because **resources are still plentiful** and need **no competition** to get,
3. **Phase C: Slowing Down Phase**
   Like the name, this is where the **growth slows down** because the **limitation starts kicking in**, such as **scarcity** (we'll discuss this in an upcoming topic).
4. **Phase D: Stationary Phase**
   The **population doesn't grow anymore** because **birth is equal to death**. This

Why is the S-Shaped Curve also called the Logistic Curve?

For me, it stands for 2 reasons:
→ It's the logic version of graph depicting the real-life situation, unlike the J-Shaped Graph that is rare in nature.
→ Logistic also means "food" or "supply". So, this graph is basically driven by the availability of food.

condition might **last for years** or the worst situation is where the **population starts to decline** due to many factors. B. Carrying Capacity Talking about the limitation that sets the population in a stagnant number, **the red dotted line** in the graph is called **carrying capacity (or simply, K)**. This is the **maximum number of individuals** being a part of the population **without destroying the environment** they're in. Simply, if a population goes **above the carrying capacity line** or the K-Line, either they will **compete for resources** or nature applies selection, the **population will eventually crash**. Environmental Brakes For Population Growth A. Density-Dependent Factors This is the type of nature brakes that will **get worse** when the population gets **too crowded**. There are three types of this density-dependent factors: → **Disease:** Especially **air-borne** diseases, they will be highly **contagious when it's crowded**. More people tend to get sick than in a small population. → **Competition:** If the **resources aren't enough** to fill the hunger, there might be a **fight** for this. → **Predation:** In animals, predators will **find and prey** more **easily on those animals in groups** than a lone one. B. Density-Independent Factors Besides those which care about how many individuals there are, here's the opposite: they're careless.	**I remember these factors by the 2 Rules:** **Density-dependent factors:** ***The more individuals you have, the harder these hit.*** **Density-independent factors:** ***We don't care who you are.***

Density-independent factors are divided into two:
→ **Natural Disasters:** A hurricane **kills** individuals **regardless of how many organisms** are there.
→ **Climate:** A **sudden freezing** drop of climate can kill the whole population, **no matter if they're crowded or not.**

Here's the summary in table:

Factors	What Is It?	Does Density Matter?
Density-Dependent	Biological limitations (food, disease)	Yes. It gets stronger as it gets crowded.
Density-Independent	Physical limitations (weather, natural disaster)	No. It hits everyone the same way.

Mentors’ Sources:

Campbell 13th edition:
[Campbell 13]